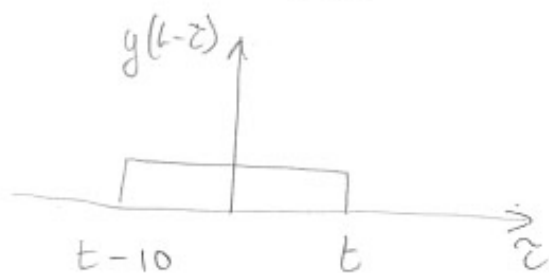


$$1) \quad y(t) = \text{rect}\left(\frac{t-5}{10}\right)$$

①



$$z(t) = \int_{t-10}^t e^{-\pi \tau^2} d\tau = \phi\left(t\sqrt{2\pi}\right) - \phi\left((t-10)\sqrt{2\pi}\right)$$

$$2) \quad x(t) = \text{triangle}\left(\frac{t-\frac{1}{2}}{\frac{1}{2}}\right) \left(\frac{e^{-j2t} e^{j\pi/6} - e^{j2t} e^{-j\pi/6}}{2j} \right)$$

$$= x_1(t) \left(\frac{e^{j\pi/6}}{2j} e^{-j\pi \frac{1}{n} t} - \frac{e^{-j\pi/6}}{2j} e^{+j\pi \frac{1}{n} t} \right)$$

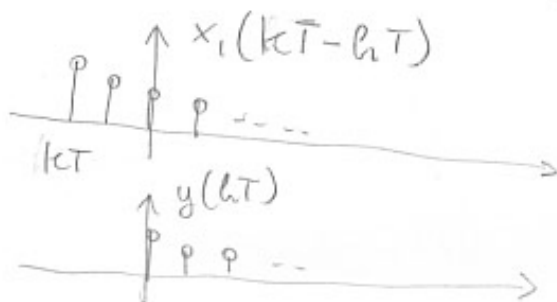
$$X_1(f) = \frac{1}{2} \text{sinc}^2\left(\frac{f}{2}\right) e^{-j\pi \frac{1}{2} f}$$

$$X(f) = \frac{e^{j\pi/6}}{2j} X_1\left(f + \frac{1}{n}\right) - \frac{e^{-j\pi/6}}{2j} X_1\left(f - \frac{1}{n}\right)$$

3)

$$z(kT) = x * y(kT) = x_1 * y(kT) + y(kT - 2T)$$

$$x_1(kT) = (0.5)^{-k} 1_0(-kT)$$



$$k \leq 0$$

$$x_1 * y(kT) = \sum_{n=0}^{+\infty} (0.5)^n (0.5)^{k-n}$$

$$= T \cdot 0.5^{-k} \sum_{n=0}^{+\infty} ((0.5)^2)^n$$

$$= T \cdot 0.5^{-k} \cdot \frac{1}{1 - 0.25}$$

(2)

$$= T \cdot 0.5^{-k} \frac{1}{1 - \frac{1}{h}} = \frac{h}{3} T \cdot 0.5^{-k}$$

per $k > 0$

$$x_1 * y(hT) = T \sum_{h=k}^{+\infty} (0.5)^h (0.5)^{h-k}$$

$$= T \sum_{h=0}^{+\infty} (0.5)^{h+k} (0.5)^h$$

$$= T \cdot 0.5^k \sum_{h=0}^{+\infty} \left(\frac{1}{4}\right)^h = \frac{h}{3} T \cdot 0.5^k$$

$$x_1 * y(hT) = \frac{h}{3} T \cdot (0.5)^{|k|}$$

h)

$$P_s = \lim_{T \rightarrow +\infty} \frac{1}{2T} \cdot \int_{-T}^T V_0^2 \cos^2\left(2\pi f_0 t + \frac{\pi}{5}\right) dt$$

$$= \lim_{k \rightarrow +\infty} \frac{1}{2kT_P} \int_{-kT_P}^{kT_P} V_0^2 \cos^2\left(2\pi f_0 t + \frac{\pi}{5}\right) dt \quad T_P = \frac{1}{f_0}$$

$$= \lim_{k \rightarrow +\infty} \frac{1}{2kT_P} \cdot 2k \int_0^{T_P} \frac{V_0^2}{2} + \frac{V_0^2}{2} \cos\left(4\pi f_0 t + \frac{2\pi}{5}\right) dt$$

$$= \frac{V_0^2}{2}$$

(3)

$$5) \quad H_T(z) = T \frac{1 - z^{-1}}{(1 - 0.2z^{-1})(1 - 0.4z^{-1})}$$

$$= T \left(\frac{4}{1 - 0.2z^{-1}} + \frac{-3}{1 - 0.4z^{-1}} \right)$$

$$h(kT) = 4 \cdot 0.2^k 1_0(kT) - 3 (0.4)^k 1_0(kT)$$